

Sentiments

Angeletos and La'O (Econometrica, 2013)

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- ➊ **Question:** can expectations move activity without changing fundamentals?
- ➋ **Model:** islands, random decentralized trade, and imperfect communication.
- ➌ **Fixed point:** local output depends on expected trading-partner output.
- ➍ **Theorem:** sentiments matter exactly when communication is imperfect.
- ➎ **Extension:** the same mechanism can be embedded in an RBC framework.

Research question

Can “animal spirits” move aggregate activity in a rational-expectations, competitive, unique-equilibrium economy with unchanged fundamentals?

What is ruled out

- preference or technology shocks;
- irrational expectations;
- sunspot equilibrium selection.

What remains

- agents are rational;
- markets are competitive;
- expectations about others' activity can differ.

The paper makes expectations of *endogenous demand* the object that sentiment shocks move.

Islands, firms, and technology

Geography and agents.

- A continuum of islands $i \in [0, 1]$.
- Each island has one representative household, one competitive firm, and one specialized good.
- In each period, island i is randomly matched with $j = m_t(i)$ for trade.

Firm on island i .

$$y_{it} = A_i n_{it}^{\theta} k_{it}^{1-\theta}, \quad 0 < \theta < 1, \quad k_{it} = K,$$

$$\pi_{it} = p_{it} y_{it} - w_{it} n_{it} - r_{it} k_{it}.$$

Crucial restriction

A_i is fixed for each island. The baseline allows fixed cross-sectional heterogeneity, but no aggregate or time-varying TFP shock.

Household preferences and the role of η

The household on island i consumes its home good c_{it} and its matched partner's good c_{it}^* :

$$U_i = \sum_{t \geq 0} \beta^t [U(c_{it}, c_{it}^*) - V(n_{it})],$$

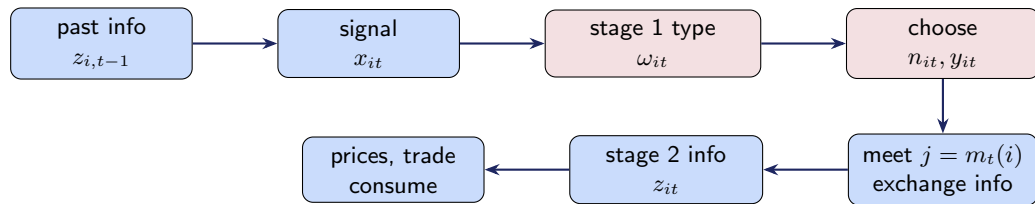
$$U(c, c^*) = \left(\frac{c}{1-\eta} \right)^{1-\eta} \left(\frac{c^*}{\eta} \right)^{\eta}, \quad V(n) = \frac{n^\epsilon}{\epsilon}, \quad \epsilon > 1.$$

$$p_{it}c_{it} + p_{it}^*c_{it}^* \leq w_{it}n_{it} + r_{it}K + \pi_{it}.$$

Interpretation of $\eta \in (0, 1)$

η is the expenditure share on the partner's good. Higher η means stronger dependence on trade and stronger exposure to expected foreign demand.

Information flow and timing



Stage 1: before meeting

$$\omega_{it} = (z_{i,t-1}, x_{it}), \quad z_{i,-1} = A_i.$$

Island i chooses employment and output using only ω_{it} .

Stage 2: after meeting

$$z_{it} = (\omega_{it}, \omega_{m_t(i),t}).$$

The matched islands learn each other's stage-1 information and trade.

Stage 2: trade and terms of trade

Normalize the household multiplier on the budget constraint to one. The consumption FOCs are

$$U_c(c_i, c_i^*) = p_i, \quad U_{c^*}(c_i, c_i^*) = p_i^*.$$

Cobb–Douglas preferences imply fixed expenditure shares. With trade balance and market clearing in the matched pair (i, j) ,

$$\begin{aligned} c_i &= (1 - \eta)y_i, & c_i^* &= \eta y_j, \\ p_i &= y_i^{-\eta} y_j^{\eta}, & p_i^* &= y_i^{1-\eta} y_j^{\eta-1}. \end{aligned}$$

Hence island i 's terms of trade are

$$R_i \equiv \frac{p_i}{p_i^*} = \frac{y_j}{y_i} = p_i^{1/\eta}.$$

If island j produces more, demand for island i 's good is higher; this is the terms-of-trade source of strategic complementarity.

Stage 1: labor and the fixed point

Production is chosen before p_{it} is observed, so the firm uses the distribution of possible partners.

$$P_t(\omega' \mid \omega) = \Pr\{\text{partner has type } \omega' \mid \omega_{it} = \omega\}.$$

Labor supply and labor demand satisfy

$$V'(n_{it}) = w_{it}, \quad w_{it} = \mathbb{E}_{it}[p_{it}] \frac{\theta y_{it}}{n_{it}}.$$

Define $\vartheta \equiv \theta/\epsilon \in (0, 1)$. Combining labor choice, technology, and expected prices gives

$$y_t(\omega) = \left(\theta \vartheta A_t(\omega) K^{1-\theta} \right)^{\frac{1}{1-\vartheta}} \left(\int_{\Omega_t} p_t(\omega, \omega') P_t(\omega' \mid \omega) d\omega' \right)^{\frac{\vartheta}{1-\vartheta}},$$

$$p_t(\omega, \omega') = y_t(\omega)^{-\eta} y_t(\omega')^{\eta}.$$

Equation (8): the core fixed point

From stage 2, $p_i = y_i^{-\eta} y_j^\eta$. Since y_i is chosen by island i ,

$$\mathbb{E}_i[p_i] = y_i^{-\eta} \mathbb{E}_i[y_j^\eta].$$

Taking logs and collecting terms yields the paper's equation (8):

$$\log y_{it} = (1 - \alpha) f_i + \alpha \tilde{\mathbb{E}}_{it}[\log y_{jt}],$$

$$f_i = \frac{1}{1 - \vartheta} \log(\theta \vartheta A_i K^{1-\theta}), \quad \alpha = \frac{\eta \vartheta}{1 - \vartheta + \eta \vartheta} = \frac{\eta}{\eta + (1 - \vartheta)/\vartheta}.$$

- f_i is island i 's fundamental log-output capacity after optimal labor adjustment.
- α is the strategic-complementarity weight: higher α means expected partner output matters more.
- $\log y_{it}$ depends on own fundamentals, $(1 - \alpha)f_i$, and expected partner activity, $\alpha \tilde{\mathbb{E}}_{it}[\log y_{jt}]$.

$\eta \uparrow \Rightarrow \alpha \uparrow$ (more dependence on foreign demand) $\theta \uparrow \Rightarrow \alpha \uparrow$ (share of labor increases) $\epsilon \downarrow \Rightarrow \alpha \uparrow$
(more elastic labor supply)

Perfect communication eliminates sentiments

Aggregate realized activity:

$$\log Y_t \equiv \int_0^1 \log y_{it} di = \kappa + \frac{1}{1-\vartheta} \bar{a}, \quad \bar{a} = \int \log A dF_A(A), \quad \kappa = \frac{1}{1-\vartheta} \log(\theta \vartheta K^{1-\theta}).$$

Aggregate beliefs:

$$\log B_t \equiv \int_0^1 \tilde{\mathbb{E}}_{it}[\log y_{m_t(i),t}] di.$$

Perfect communication gives $\tilde{\mathbb{E}}_{it}[\log y_{m_t(i),t}] = \log y_{m_t(i),t}$, so

$$\log B_t = \int_0^1 \log y_{m_t(i),t} di = \int_0^1 \log y_{it} di = \log Y_t.$$

The value of Y_t follows from adding the two equations in each match:

$$\log y_i = (1-\alpha)f_i + \alpha \log y_j, \quad \log y_j = (1-\alpha)f_j + \alpha \log y_i \quad \Rightarrow \quad \log y_i + \log y_j = f_i + f_j,$$

Average beliefs coincide with average realized activity.

Imperfect communication: Gaussian sentiment example

To show sufficiency, the paper constructs a Gaussian information structure. Normalize $K = 1$ and let

$$\log A_i \sim N(0, \sigma_A^2), \quad \xi_t \sim N(0, \sigma_\xi^2) \text{ i.i.d. over time.}$$

Here ξ_t is the common sentiment shock: it moves beliefs without changing fundamentals. Island i receives two signals about its trading environment:

$$x_{it}^1 = \log A_j + u_{it}^1, \quad x_{it}^2 = x_{jt}^1 + \xi_t + u_{it}^2.$$

First signal

x_{it}^1 is noisy information about partner productivity:

$$A_j \uparrow \Rightarrow y_j \uparrow \Rightarrow \text{demand for } i \uparrow.$$

Second signal

x_{it}^1 is what i hears about j . $x_{jt}^2 = x_{jt}^1 + \xi_t + u_{jt}^2$ is what j hears about i 's information. Likewise, x_{it}^2 tells i about j 's information.

Reading a high x_{it}^2

If i observes high $x_{it}^2 = x_{jt}^1 + \xi_t + u_{it}^2$, then either x_{jt}^1 is probably high, so j expects i to produce more and raises y_j through $\log y_j = (1 - \alpha)f_j + \alpha\mathbb{E}_j[\log y_i]$, or ξ_t is high. Either way, high x_{it}^2 predicts higher y_j .

Proposition 3: sentiment moves output and optimism

In the Gaussian example, equilibrium output is linear in local fundamentals and signals:

$$\log y_{it} = \phi_0 + \phi_a \log A_i + \phi_1 x_{it}^1 + \phi_2 x_{it}^2, \quad \phi_a > 0, \quad \phi_1 > 0, \quad \phi_2 > 0.$$

Expected prices satisfy

$$\mathbb{E}_i[\log p_{it}] = \psi_0 - \psi_a \log A_i + \psi_1 x_{it}^1 + \psi_2 x_{it}^2, \quad \psi_a, \psi_1, \psi_2 > 0.$$

Aggregation

The common component ξ_t survives aggregation, while idiosyncratic noises wash out:

$$\log Y_t = \chi_0 + \phi_2 \xi_t, \quad \log B_t = b_0 + b_\xi \xi_t, \quad \phi_2, b_\xi > 0.$$

$$\xi_t \uparrow \Rightarrow x_{it}^2 \uparrow \Rightarrow \mathbb{E}_i[y_j] \uparrow \Rightarrow \mathbb{E}_i[p_i] \uparrow \Rightarrow n_i \uparrow \Rightarrow y_i \uparrow.$$

$$\begin{aligned} \mathbb{E}_i[p_i] = y_i^{-\eta} \mathbb{E}_i[y_j^\eta] &\Rightarrow \mathbb{E}_i[y_j^\eta] \uparrow \Rightarrow \mathbb{E}_i[p_i] \uparrow, \\ n_i = \left[\theta A_i K^{1-\theta} \mathbb{E}_i[p_i] \right]^{1/(\epsilon-\theta)} &\Rightarrow \mathbb{E}_i[p_i] \uparrow \Rightarrow n_i \uparrow. \end{aligned}$$

Conclusion of the baseline theory

- ① **Methodological move:** decentralized trade limits the communication normally embedded in markets.
- ② **Core equation:** output responds to expected trading-partner output, $\tilde{\mathbb{E}}_i[\log y_j]$.
- ③ **Sharp theorem:** Y_t and B_t respond to ξ_t if and only if communication is imperfect.
- ④ **Equilibrium discipline:** sentiments move beliefs and behavior along a unique rational-expectations equilibrium.

Next extensions in the paper

Higher-order belief shocks collapse into a composite sentiment; repeated communication generates contagion and boom–bust dynamics; the mechanism can be embedded in an RBC-style quantitative model.

RBC extension: what changes?

The quantitative section keeps the sentiment mechanism but places it inside a richer business-cycle model.

Standard RBC logic

Aggregate technology shocks move marginal products, income, employment, consumption, and investment.

Specialized goods become intermediate inputs into a final good:

$$q_{it} = A_i(e_{it}k_{it})^{1-\theta}n_{it}^{\theta}, \quad y_{it} = \frac{1}{\zeta}h_{it}^{1-\eta}(h_{it}^*)^{\eta}.$$

Final output is used for consumption and investment:

$$c_{it} + i_{it} = y_{it}, \quad k_{i,t+1} = (1 - \Delta(e_{it}))k_{it} + i_{it}, \quad \Delta(e) = \frac{\delta}{\mu}e^{\mu}.$$

This paper's logic

There is no aggregate TFP shock. A sentiment shock moves beliefs about demand from other islands.

The same expected-demand channel now moves employment, utilization, consumption, and investment.

RBC calibration and data

The computational shortcut replaces the conventional TFP shock with a persistent sentiment state:

$$x_{it} = \log A_j + \varepsilon_{it}, \quad \xi_t = \rho \xi_{t-1} + \nu_t, \quad \tilde{K}_{t+1} = \Gamma(\xi_t, \tilde{K}_t).$$

Parameters

- quarterly frequency;
- $\beta = 0.99$, $\gamma = 2$, $\theta = 0.65$, $\epsilon = 2$, $\mu = 2$;
- $\rho = 0.98$, $\eta = 1$;
- $\sigma_A = \sigma_\varepsilon = \sigma_\xi = 0.038$ to match output volatility.

Simulation and data

- 1000 simulated series, each 1250 quarters;
- drop first 1000 quarters, HP-filter last 250 with weight 1600;
- U.S. quarterly data, 1948Q3–2012Q2, seasonally adjusted.

RBC quantitative results

Variable	Model		U.S. data	
	s.d.	corr. with Y	s.d.	corr. with Y
Y	1.67	1.00	1.67	1.00
N	1.41	1.00	1.43	0.87
C	1.21	0.98	1.27	0.80
I	4.14	0.96	5.48	0.82
Y/N	0.26	0.99	0.82	0.51
LW	4.95	-1.00	4.98	-0.82

All series are quarterly and HP-filtered; output volatility is matched by construction.

- Sentiments generate procyclical employment, consumption, investment, and labor productivity.
- The model under-generates labor-productivity volatility because aggregate technology shocks are absent.
- It matches the countercyclical labor wedge: pessimism looks like weak aggregate demand.

Thank you

Questions